Unit-3

Set-1:

1. Which of the following benefits is associated with low coupling?

1 Marks

- Increased cohesion

Facilitates parallel development

Increased complexity

Tightly coupled code

2. The design should be a readable, Understandingable guide for those who generate _____.

1 Marks

- requirements

tests

code

models

3. Which design approach emphasizes separation of tasks into smaller, reusable functional units?

1 Marks

- Object-Oriented Design

Component-Level Design

Behavioral Design

Function-Oriented Design

4. Which combination of models provides a complete software design representation?

1 Marks

- Coding, Testing, Deployment, Maintenance

Abstraction, Encapsulation, Cohesion, Polymorphism

Modules, Interfaces, Patterns, Frameworks

Data/Class, Architectural, Interface, Component-Level

5. Content design in WebApps involves structuring _____.

1 Marks

- Icons

Content objects

System hardware

Navigation menus

6. Object-oriented design is based on real-world entities called _____.

1 Marks

- functions

modules

components

objects

7. Golden Rule #1 of UI Design is _____.

1 Marks

- **Place the user in control**

Reduce the user's memory load

Make the interface consistent

Improve performance

8. Which design element includes state and sequence diagrams?

1 Marks

- Interface Design

Component-Level Design

Architectural Design

Behavioral Design

9. Meaningful defaults help users by _____.

1 Marks

- Increasing memory load

Reducing setup effort

Decreasing consistency

Hiding options

10. Which benefit of low coupling allows different teams to work on separate modules simultaneously?

1 Marks

- Improved Maintainability

Facilitates Parallel Development

Better Scalability

Improved Testability

11. High cohesion modules have a clear _____ and fewer dependencies.

1 Marks

- **focus**

complexity

size

cost

12. Which is NOT a benefit of high cohesion?

1 Marks

- Improved maintainability

Enhanced modularity

Increased complexity

Increased reusability

13. Which design enhances visual appeal of a WebApp?

1 Marks

- Content Design

Aesthetic Design

Navigation Design

Architecture Design

14. Which of the following BEST explains why high cohesion improves reusability?

1 Marks

- Because modules serve multiple unrelated tasks

Because modules have fewer classes

Because modules focus on a single well-defined purpose

Because modules are loosely connected

15. Low coupling promotes _____, allowing developers to build, test, and maintain modules independently.

1 Marks

- cohesion

coupling

modularity

scalability

16. Which attribute of FURPS considers human factors and consistency?

1 Marks

- Functionality

Reliability

Supportability

Usability

17. Golden Rule #3 of UI Design is _____.

1 Marks

- Place the user in control

Reduce memory load

Make the interface consistent

Enhance modularity

18. Navigation design defines how users _____.

1 Marks

- Modify system code

Move through a WebApp

Enhance visuals

Store data

19. A design should be derived using a _____ method.

1 Marks

- Random

Approximate

Flexible

Repeatable

20. If designing a payment WebApp, which principle best ensures familiar user experience?

1 Marks

- Undo

Use real-world metaphor

Hide internals

Flexible interaction

21. Which of the following is a benefit of low coupling related to making changes without affecting the entire system?

1 Marks

- **Improved Maintainability**

Improved Testability

Better Code Organization

High Cohesion

22. Which domain must be addressed to provide a complete picture of software?

1 Marks

- Only technical

Only business

Deployment

Data, functional, and behavioral

23. To give users freedom, software should allow _____ actions.

1 Marks

- Redo

Undo

Repeat

Skip

24. Loosely coupled systems are more _____ to change, making it easier to add or remove modules.

1 Marks

- **adaptable**

rigid

dependent

complex

25. Which attribute of FURPS is measured using speed, throughput, and efficiency?

1 Marks

- Usability

Reliability

Functionality

Performance

26. Interfaces among modules identify exact _____ exchanged.

1 Marks

- algorithms

modules

requirements

data items

27. A _____ is a modular building block for computer software.

1 Marks

- function

object

component

class

28. A design should lead to _____ that are appropriate for the classes.

1 Marks

- patterns

styles

data structures

frameworks

29. Which principle is ensured by component-level design along with reusability and modularity?

1 Marks

- Inheritance

Abstraction

Performance

Maintainability

30. Lower coupling means that modules are more _____.

1 Marks

- dependent

connected

complex

independent

set-2:

1. Why should consistency not be broken unless necessary?

1 Marks

- It enhances memory load

It breaks user expectations

It improves performance

It reduces flexibility

2. To give users freedom, software should allow _____ actions.

1 Marks

- Redo

Undo

Repeat

Skip

3. Low coupling encourages a more _____ codebase, with each module having clearly defined responsibilities.

1 Marks

- **organized**

independent

monolithic

fragmented

4. A design should be _____; the software should be partitioned into elements or subsystems.

1 Marks

- dynamic

linear

modular

scalable

5. Which of the following is NOT one of the four key models of software design?

1 Marks

- Data/Class Design

Architectural Design

Deployment Design

Behavioral Design

6. The CRC diagram stands for _____.

1 Marks

- Class-Relation-Component

Component-Relation-Collaborator

Class-Responsibility-Collaborator

Class-Requirement-Collaboration

7. Low coupling encourages a more _____ codebase, with each module having clearly defined responsibilities.

1 Marks

- complex

organized

independent

scalable

8. Golden Rule #2 of UI Design is _____.

1 Marks

- Place the user in control

Reduce the user's memory load

Make the interface consistent

Increase flexibility

9. Loosely coupled systems are more _____ to change, making it easier to add or remove modules.

1 Marks

- **adaptable**

rigid

dependent

complex

10. Function-oriented design focuses on _____ that perform operations on data.

1 Marks

- components

objects

interfaces

functions/procedures

11. Which of the following is a focus of component-level design?

1 Marks

- **Testing components**

Deploying modules

Documenting processes

Structuring individual components

12. The _____ of a system or product must be represented during design steps.

1 Marks

- interface

component

design

architecture

13. Cohesion refers to the _____ to which elements within a module are related and work together.

1 Marks

- **degree**

independence

complexity

adaptability

14. Which principle ensures software is well-structured, efficient, and maintainable?

1 Marks

- Design patterns

Design modules

Design concepts

Design principles

15. Software design translates requirements into a _____ for construction.

1 Marks

- module

diagram

blueprint

pattern

16. Component-level design acts as a bridge between system design and _____.

1 Marks

- deployment

testing

coding

maintenance

17. Which combination is ideal for software design?

1 Marks

- High coupling & High cohesion

Low coupling & High cohesion

Low cohesion & High coupling

High coupling & Low cohesion

18. Which characteristic ensures a design is logically partitioned?

1 Marks

- Usability

Performance

Reliability

Modularity

19. A design should lead to _____ that are appropriate for the classes.

1 Marks

- patterns

styles

data structures

frameworks

20. Navigation design defines how users _____.

1 Marks

- Modify system code

Move through a WebApp

Enhance visuals

Store data

21. High cohesion and low coupling together lead to a system that is more _____, maintainable, and reusable.

1 Marks

- **scalable**

fragile

complex

rigid

22. Content design in WebApps involves structuring _____.

1 Marks

- Icons

Content objects

System hardware

Navigation menus

23. Different _____ are required to implement the design solution.

1 Marks

- algorithms

data structures

modules

interfaces

24. Which design enhances visual appeal of a WebApp?

1 Marks

- Content Design

Aesthetic Design

Navigation Design

Architecture Design

25. If an app family (e.g., MS Office) shares same UI design, it reflects _____.

1 Marks

- Undo action

Consistency across product line

Progressive disclosure

Aesthetic design

26. Design allows you to model the _____ or product that is to be built.

1 Marks

- design

data

system

process

27. Which pyramid level focuses on handling dynamic content and data?

1 Marks

- Navigation

Component Design

Interface

Aesthetic

28. Which of the following is directly improved by low coupling in terms of testing?

1 Marks

- Improved Maintainability

Improved Testability

Better Scalability

Increased Flexibility

29. High cohesion improves _____, making it easier to modify or replace components without affecting the system.

1 Marks

- **maintainability**

scalability

coupling

organization

30. Which benefit is shared by both low coupling and high cohesion?

1 Marks

- **Improved maintainability**

Increased complexity

Parallel execution

Loose structure

Set-3:

1. Which design describes how software communicates with systems and humans?

1 Marks

- Architectural Design

Data/Class Design

Component-Level Design

Interface Design

2. Design concepts provide the _____ for developing robust and maintainable software.

1 Marks

- guideline

framework

foundation

pattern

3. Lower coupling means that modules are more _____.

1 Marks

- dependent

connected

complex

independent

4. A design should be _____; the software should be partitioned into elements or subsystems.

1 Marks

- dynamic

linear

modular

scalable

5. Which design approach uses UML Class Diagrams and Sequence Diagrams?

1 Marks

- Function-Oriented Design

Component-Level Design

Coupling Design

Object-Oriented Design

6. The CRC diagram stands for _____.

1 Marks

- Class-Relation-Component

Component-Relation-Collaborator

Class-Responsibility-Collaborator

Class-Requirement-Collaboration

7. High cohesion improves _____, making it easier to modify or replace components without affecting the system.

1 Marks

- **maintainability**

scalability

coupling

organization

8. Component-level design acts as a bridge between system design and _____.

1 Marks

- deployment

testing

coding

maintenance

9. Function-oriented design focuses on _____ that perform operations on data.

1 Marks

- components

objects

interfaces

functions/procedures

10. Which combination is ideal for software design?

1 Marks

- High coupling & High cohesion

Low coupling & High cohesion

Low cohesion & High coupling

High coupling & Low cohesion

11. Golden Rule #2 of UI Design is _____.

1 Marks

- Place the user in control

Reduce the user's memory load

Make the interface consistent

Increase flexibility

12. Golden Rule #1 of UI Design is _____.

1 Marks

- **Place the user in control**

Reduce the user's memory load

Make the interface consistent

Improve performance

13. What helps in formulating a product or system?

1 Marks

- Code

Testing

Documentation

Design Concepts

14. Without design, the system may become _____.

1 Marks

- Stable

Robust

Maintainable

Unstable

15. Which element of software design starts after requirements are analyzed?

1 Marks

- Testing

Design

Deployment

Design

16. A design should be derived using a _____ method.

1 Marks

- Random

Approximate

Flexible

Repeatable

17. A design should lead to _____ that are appropriate for the classes.

1 Marks

- patterns

styles

data structures

frameworks

18. Low coupling encourages a more _____ codebase, with each module having clearly defined responsibilities.

1 Marks

- complex

organized

independent

scalable

19. High cohesion facilitates development and _____ of modules in isolation.

1 Marks

- **testing**

debugging

execution

compilation

20. Which two steps achieve software design goals?

1 Marks

- Modeling and Testing

Coding and Deployment

Debugging and Documentation

Diversification and Convergence

21. Which domain must be addressed to provide a complete picture of software?

1 Marks

- Only technical

Only business

Deployment

Data, functional, and behavioral

22. High cohesion modules have a clear _____ and fewer dependencies.

1 Marks

- **focus**

complexity

size

cost

23. The pyramid base in WebApp design represents _____.

1 Marks

- Interface design

Aesthetic design

Technology/Component Design

Navigation design

24. Which of the following provides fundamental guidelines or best practices to shape software?

1 Marks

- Design patterns

Design concepts

Design principles

Modules

25. High cohesion and low coupling together lead to a system that is more _____, maintainable, and reusable.

1 Marks

- **scalable**

fragile

complex

rigid

26. Function-oriented design uses a _____ model such as DFDs.

1 Marks

- object

class

data flow

state

27. One fundamental characteristic of software design is that it should implement all _____ requirements.

1 Marks

- implicit

explicit

abstract

derived

28. Which principle allows advanced users to customize interactions?

1 Marks

- Undo

Streamline interaction as skill levels advance

Consistency across product

Progressive disclosure

29. If designing a payment WebApp, which principle best ensures familiar user experience?

1 Marks

- Undo

Use real-world metaphor

Hide internals

Flexible interaction

30. Progressive disclosure presents information _____.

1 Marks

- All at once

In stages

Randomly

Never

Set-4:

1. A design should be derived using a _____ method.

1 Marks

- Random

Approximate

Flexible

Repeatable

2. FURPS quality attributes were developed by _____.

1 Marks

- IBM

Microsoft

Hewlett-Packard

Google

3. Content design in WebApps involves structuring _____.

1 Marks

- Icons

Content objects

System hardware

Navigation menus

4. Stakeholder requirements, business needs, and _____ considerations come together in design concepts.

1 Marks

- design

economic

technical

legal

5. Which design describes how software communicates with systems and humans?

1 Marks

- Architectural Design

Data/Class Design

Component-Level Design

Interface Design

6. Meaningful defaults help users by _____.

1 Marks

- Increasing memory load

Reducing setup effort

Decreasing consistency

Hiding options

7. The CRC diagram stands for _____.

1 Marks

- Class-Relation-Component

Component-Relation-Collaborator

Class-Responsibility-Collaborator

Class-Requirement-Collaboration

8. Which of the following benefits is associated with low coupling?

1 Marks

- Increased cohesion

Facilitates parallel development

Increased complexity

Tightly coupled code

9. Which benefit is shared by both low coupling and high cohesion?

1 Marks

- **Improved maintainability**

Increased complexity

Parallel execution

Loose structure

10. Loosely coupled systems are more _____ to change, making it easier to add or remove modules.

1 Marks

- **adaptable**

rigid

dependent

complex

11. If a system has low cohesion and high coupling, what is the most likely outcome?

1 Marks

- The system becomes highly modular and reusable.

The system becomes harder to maintain, test, and reuse.

The system becomes more flexible and scalable.

The system eliminates unintended consequences.

12. Function-oriented design focuses on _____ that perform operations on data.

1 Marks

- components

objects

interfaces

functions/procedures

13. High cohesion modules have a clear _____ and fewer dependencies.

1 Marks

- **focus**

complexity

size

cost

14. Without design, the system may become _____.

1 Marks

- Stable

Robust

Maintainable

Unstable

15. Progressive disclosure presents information _____.

1 Marks

- All at once

In stages

Randomly

Never

16. Which principle enables undo functionality for better control?

1 Marks

- Consistency

Allow user interaction to be interruptible

Flexible interaction

Parallel design

17. Supportability in FURPS combines which factors?

1 Marks

- Performance, Efficiency, Recovery

Consistency, Modularity, Reliability

Scalability, Testing, Portability

Extensibility, Adaptability, Serviceability

18. High cohesion improves _____, making it easier to modify or replace components without affecting the system.

1 Marks

- **maintainability**

scalability

coupling

organization

19. Which combination is ideal for software design?

1 Marks

- High coupling & High cohesion

Low coupling & High cohesion

Low cohesion & High coupling

High coupling & Low cohesion

20. Low coupling enables developers to make changes to individual modules without _____ the entire system.

1 Marks

- breaking

disrupting

restarting

compiling

21. Function-oriented design uses a _____ model such as DFDs.

1 Marks

- object

class

data flow

state

22. Which attribute of FURPS considers human factors and consistency?

1 Marks

- Functionality

Reliability

Supportability

Usability

23. Navigation design defines how users _____.

1 Marks

- Modify system code

Move through a WebApp

Enhance visuals

Store data

24. Which UI design principle hides unnecessary technical details from casual users?

1 Marks

- Consistency

Hide technical internals

Aesthetic design

Navigation design

25. Which is NOT a benefit of high cohesion?

1 Marks

- Improved maintainability

Enhanced modularity

Increased complexity

Increased reusability

26. Cohesion refers to the _____ to which elements within a module are related and work together.

1 Marks

- **degree**

independence

complexity

adaptability

27. Design allows you to model the _____ or product that is to be built.

1 Marks

- design

data

system

process

28. Diversification means gathering a wide range of _____ and solutions.

1 Marks

- requirements

components

data

ideas

29. The pyramid base in WebApp design represents _____.

1 Marks

- Interface design

Aesthetic design

Technology/Component Design

Navigation design

30. Which of the following is a focus of component-level design?

1 Marks

- **Testing components**

Deploying modules

Documenting processes

Structuring individual components